

Oway: Meeting customers' needs while reducing business costs during COVID-19

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

ABOUT Oway

Oway moved its ride-sharing platform to Google Cloud to control cloud costs, match infrastructure resources with the changing business demands and improve decision-making time analytics.



About Oway

Oway is a pioneer in Myanmar's tech ecosystem, having driven consumer adoption of mobile apps with its travel web portal Oway Travel and ride-sharing service Oway Ride. The technology company also provides corporate transportation services with Oway Fleet. Oway continues to innovate and expand its offering with new products such as Oway Fresh, an online grocery platform. In 2020, it was nominated Myanmar's leading travel agency in the World Travel Awards.

Google Cloud results

- Reduces 30% of the time spent setting up new servers by autoscaling on Compute Engine
- Accelerates data analysis on Cloud SQL to improve decision-making and customer experience
- Deploys remote cloud migration without service disruption in six weeks
- Improves location accuracy of

Saves cloud : pay-as billing

Industries: Transportation, Technology, Travel & Hospitality

Location: Myanmar

esources/analytics-product-overview/)

virtusa

About Virtusa

Virtusa is a global IT service provider that offers end-to-end cloud services to customers in their cloud journey, from strategy and planning to migration and managed services. Virtusa gained Google Cloud Infrastructure Partner Specialization for its proven expertise and success in assisting clients to architect and build their Google Cloud infrastructure and workflows and migrate to Google Cloud.

ation, direction, strategy

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud SQL (<https://cloud.google.com/sql/>) help.

Google Maps Platform

(<https://mapsplatform.google.com/>)

Google Analytics

(<https://marketingplatform.google.com/about/resources/analytics-product-overview/>)

virtusa

ride-hailing
service
with
Google
Maps APIs

Migrate to Virtual Machines

(<https://cloud.google.com/migrate/virtual-machines>)

Virtusa is a global IT service provider that

Geocoding API

(<https://developers.google.com/maps/documentation/geocoding/intro>)

strategy and planning to migration and

Maps iOS SDK

(<https://developers.google.com/maps/documentation/ios-sdk/intro>)

for its proven expertise and success in

Maps Android SDK

(<https://developers.google.com/maps/documentation/android-sdk/intro>)

workflows and migrate to Google Cloud

Google Places API

(<https://cloud.google.com/maps-platform/places/>)

Google Maps API

Distance Matrix

(<https://developers.google.com/maps/documentation/distance-matrix/start>)

Google Maps API

Directions API

(<https://developers.google.com/maps/documentation/directions/start>)

From news and information to find more, today 22 million

(<https://datareportal.com/reports/digital-internet-users-in-myanmar-rely-on-google>)

internet users in Myanmar rely on Google what they're looking for. But amid

economy.

(<https://oxfordbusinessgroup.com/insights/2020/economy#:~:text=From%20The%20an%202020,youthful%20population%20its%20progress.>)

, there's still a large untapped space

transportation sector. Myanmar's first

agency appeared as recently as 2017

launch of Oway Travel (<https://owaytravel.com>)

the Oway group. After working in California for seven

years, the company's founder, Nay Aung, decided to

move back home to develop a series of internet-

based ventures. Realizing that there was a need for

people in Myanmar to be able to book travel plans

digitally just as easily as they could do anything else

online, he launched Oway Travel, followed by the

country's first ride-hailing agency, Oway Ride, and

Oway Fleet for corporate vehicle leasing.

Oway's travel and ride-hailing businesses was a

successful venture, but like many businesses around

the world, it experienced unforeseen challenges due

to COVID-19. While Oway's leasing operation was

less impacted, the fact that most people were

staying at home as a quarantine measure meant that fewer people were using Oway's app to book a trip. The company noticed that instead, more people were looking for ways to have food delivered to their homes. Responding quickly to the shift in consumer needs, Oway responded by launching an online grocery business called Oway Fresh.

It was a smart move by a company with its finger to the pulse of what people in Myanmar want, and Oway wanted to make sure it was agile enough to keep responding as needs changed. The solution? A scalable infrastructure that would allow it to match cloud capacity to demand. By doing this, it would be able to reduce unnecessary costs and reinvest the savings into other projects and areas of its growing business. To bring this strategy to life, in May 2020, the company began moving its workload to Google Cloud (<https://cloud.google.com/>), starting with Oway Ride.

“Our goal is to
build a scalable
platform on
Google Cloud,
with on-demand
capacity for any
workload.
Automated

features such as
autoscaling on
Compute Engine
help us to set up
new servers 30%
faster, so we can
accommodate
traffic spikes
while focusing
on modernizing
our
infrastructure for
long-term
growth.”

—Tun Tun Linn, Group
Chief Technology Officer,
Oway Company Limited

Staying agile and resilient with efficient use of cloud resources

With its previous infrastructure solution, Oway paid in advance for access to a vast cloud resource at a discounted price. However, with most people in Myanmar under self-quarantine, it saw up to an 80% dip in traffic on Oway’s travel and ride-hailing services, which meant that the company was paying for extra resources that it didn’t always need to call upon.

Oway quickly realized that these cloud costs would be better reallocated to other areas of the business, helping to fund development for the needs of the moment. But with the previous infrastructure, there was no way to adapt to changing resource needs.

With Google Cloud as its new cloud solution, Oway has gained the flexibility that it needs. It uses [Compute Engine](#) (/compute) to host its application servers, including [Cloud SQL](#) (/sql) as its managed database service and predictive analytics on Google Cloud. It also uses [Google Maps Platform](#) (/maps-platform) as its navigation solution for Oway Ride and Oway Fleet app users, that includes delivery drivers for Oway Fresh.

“Our goal is to build a scalable platform on Google Cloud, with on-demand capacity for any workload,” says Tun Tun Linn, Group Chief Technology Officer at Oway. “Automated features such as autoscaling on Compute Engine help us to set up new servers 30% faster, so we can accommodate traffic spikes while focusing on modernizing our infrastructure for long-term growth.”

By controlling fixed expenses during a time of downturn, Oway has also improved cash flow, ensuring that the business can stay agile and resilient during COVID-19 and beyond. “Google Cloud keeps our business running smoothly for customers without making drastic cost cuts to our operations,” says Tun Tun. “The scalable platform gives us the flexibility to spin up servers and increase capacity when the market rebounds. In the

meantime, its pay-per-use model reduces our cloud expenses by 20%.”

“Data insights from Google Analytics and Cloud SQL can help us make smarter and faster decisions to run our business. With real-time visibility of our performance, we can make timely decisions to adjust to changing market conditions. It’s impossible to be agile if we’re looking at month-old data.”

—Tun Tun Linn, Group
Chief Technology Officer,

Aligning the cloud migration strategy with business goals

For a successful migration, Oway turned to Google Cloud Premier Partner [Virtusa](https://www.virtusa.com/)

(<https://www.virtusa.com/>) for its experience in helping businesses to migrate, architect, and build their Google Cloud infrastructure and workflows. Virtusa worked with Oway to formulate a detailed migration strategy, from assessing workloads and preparing the cloud framework, to supporting the live migration and optimizing the infrastructure post-migration. As part of the evaluation process, the migration team carried out a week-long proof-of-concept (POC) with its shortlisted cloud providers to test performance requirements such as availability and reliability.

“We chose Google Cloud for its synergy with our business,” says Tun Tun. “Google’s native support for Kubernetes is aligned with our technology roadmap to fully containerize our platform. And because we rely heavily on Google Maps Platform for Oway Ride, it makes sense to host the ride-hailing platform on Google Cloud for seamless integration.”

Oway and Virtusa worked closely to execute a smooth migration to minimize downtime for customers. In the weeks following the POC, the migration team deployed Oway Ride in a staging

environment, before moving it to a live production during a two-hour maintenance window.

COVID-19 presented a logistical challenge to Oway since travel restrictions prevented the team from being onsite for the live migration. Although cloud-to-cloud migrations don't involve physical hardware, administrators need to coordinate in real time, as if they are sitting next to each other, to troubleshoot unforeseen disruption to the migration. As such, the team kept in close contact throughout the live migration. Thankfully, no adjustments were required. Oway Ride app users experienced no issues logging into the app on the new infrastructure the following day.

"You need a strong and knowledgeable partner for remote cloud migration to avoid costly mistakes such as data loss, and thankfully, Virtusa made the migration a seamless one," says Tun Tun. Virtusa worked alongside Oway's cloud engineers to share hands-on Google Cloud experience so it could carry out the migration smoothly and optimize the app's infrastructure after the migration.

"We simplified the migration process by using

Migrate to Virtual Machines

(<https://cloud.google.com/migrate/virtual-machines>), a Google Cloud migration tool, to move Oway Ride to Compute Engine," says Naresha Supramaniam, SVP, Head of APAC, Virtusa. "Using a lift-and-shift approach, Oway migrated the application to Google Cloud quickly and painlessly, through carefully planned phases, without having to rebuild the infrastructure."

“Google Cloud keeps our business running smoothly for customers without making drastic cost cuts to our operations. The scalable platform gives us the flexibility to spin up servers and increase capacity when the market rebounds. In the meantime, its pay-per-use model reduces our cloud expenses by 20%.”

*—Tun Tun Linn, Group
Chief Technology Officer,
Oway Company Limited*

Smart analytics to enhance business intelligence and user experience

From GPS information generated by its Oway Ride vehicles to inventories on Oway Fresh, Oway creates massive amounts of data across its organization. However, many of these transactions are created offline. For example, some Oway Travel customers prefer to book a flight with a live agent on the phone, instead of buying their ticket online. To gain a holistic view of its business operations, Oway is combining offline and online data on Cloud SQL to analyze key performance metrics such as revenue, profit margins, and performance for each business unit.

“Data insights from [Google Analytics](https://analytics.google.com/analytics/web/provision/#/provision)

(<https://analytics.google.com/analytics/web/provision/#/provision>)

and Cloud SQL can help us make smarter and faster decisions to run our business,” says Tun Tun. “With real-time visibility of our performance, we can make timely decisions to adjust to changing market conditions. It’s impossible to be agile if we’re looking at month-old data.”

Aiming to understand and better predict customer needs through data, as well as improve customer

experience, Oway has been exploring the machine learning capabilities of Google Cloud. In one project, it has built a predictive model to forecast demand for Oway Ride services by analyzing patterns in how and when users make bookings. Based on that information, the model alerts drivers to head toward areas where and when demand is likely to be higher, minimizing waiting time for customers.

Another predictive model built by Oway recommends hotels to users at the moment when they are booking a flight on Oway Travel, anticipating what they are likely to look for next and helping them save time and effort. Oway believes that travelers will be able to benefit from such personalized recommendations when they start traveling again.

Whether you're a tourist traveling around Myanmar or a local getting to work, Oway has built its reputation as the go-to option for ride-hailing in Myanmar. But one of the company's challenges has always been around its map accuracy, especially in rural areas. Oway experimented with different mapping solutions, but found that they lacked updated maps. In order to gain reliable, comprehensive data and comprehensive coverage, the company turned to Google Maps Platform. Today, it deploys a range of Google Maps Platform APIs and SDKs to provide accurate real-time location data on its ride-sharing app.

With the [Maps SDK for Android](https://developers.google.com/maps/documentation/android-sdk/overview)

(<https://developers.google.com/maps/documentation/android-sdk/overview>)

and [Maps SDK for iOS](#)

(<https://developers.google.com/maps/documentation/ios-sdk/overview>)

, Oway Ride displays interactive maps on any mobile device. Oway uses the [Places API](#)

(<https://developers.google.com/places/web-service/overview>)

to add points of interest that provide drivers with additional context, the [Directions API](#)

(<https://developers.google.com/maps/documentation/directions/overview>)

for real-time routing, and the [Distance API](#)

(<https://developers.google.com/maps/documentation/distance-matrix/overview>)

to calculate the distance in kilometers. Passengers can keep track of their driver's arrival time on the

app, thanks to the [Distance Matrix API](#)

(<https://developers.google.com/maps/documentation/distance-matrix/overview>)

that calculates travel time based on real-time traffic conditions.

If customers opt in to location services, the app uses reverse geocoding, a feature of the [Geocoding API](#)

(<https://developers.google.com/maps/documentation/geocoding/overview>)

, to convert their GPS coordinates to a street address in the pickup field, simplifying the booking process. The Directions API helps drivers get around more easily with turn-by-turn navigation so they can pick up more passengers more efficiently.

To further enhance map accuracy, Oway is currently working with local data collection agencies to create a library of home addresses in major cities. With this information, developers will be able to deploy map

updates in real time by converting addresses to geographical coordinates using the Geocoding API.

Preparing a foundation for COVID-19 recovery

Today, Oway runs 50% of its workload on Google Cloud. Once the migration is complete, the company plans to modernize its monolithic platform before the end of 2020, by shifting to a microservices architecture on Google Kubernetes Engine (/kubernetes-engine). This will allow it to accelerate application development and stay agile to develop new service additions and features such as loyalty programs and AI-powered recommendation engines and a more personalized user experience for its customers. Meanwhile, Oway continues to work closely with Virtusa to assess workloads in its legacy cloud environment such as Oway Travel and Oway Fresh for future migration to Google Cloud.

“The challenges of COVID-19 are there,” says Tun Tun, “but so are the opportunities to help more people in Myanmar get what they need during this time. To that end, we’re committed to providing our customers with key services such as grocery delivery. The switch to our new infrastructure on Google Cloud has been timely, helping us to gain cost efficiencies when most needed so that we can work faster and better for our customers and, despite the current challenges, become a stronger business.”